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STORAGE TUB WITH ACCESS PANEL FOR A VEHICLE

Abstract

A vehicle component includes a tub configured to store cargo and an access panel configured to cover at least a portion of the bay defined by the vehicle. A living hinge couples the tub to the access panel. All of the tub, the access panel, and the living hinge are monolithically formed in a common mold, such as using gate sequenced injection molding. The component may include side portions secured to the tube. The living hinge may include a first portion connecting a first side portion to the access panel and a second portion connecting a second side portion to the access panel. The tub may be detached from the access panel between the first and second side portions. An upper edge of the tube may have a dust seal secured thereto that engages a cover, e.g., hood, configured to cover the bay.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] This application claims priority to and benefit of U.S. Provisional Patent Application No. 63/558,920, filed Feb. 28, 2024, which is assigned to the assignee hereof and hereby expressly incorporated herein in its entirety as if fully set forth below and for all applicable purposes.

INTRODUCTION

[0002] The present disclosure relates to a storage tub with an integrated access panel for a vehicle.

SUMMARY

[0003] The present disclosure describes an approach for implementing an access panel in a bay of a vehicle. In one aspect, a vehicle component includes a tub configured to store cargo and an access panel configured to cover at least a portion of the bay defined by the vehicle. A living hinge couples the tub to the access panel. All of the tub, the access panel, and the living hinge are monolithically formed in a common mold.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0004] FIG. 1A illustrates an example vehicle that may be used in accordance with certain embodiments.

[0005] FIG. 1B illustrates a chassis of a vehicle having multiple drive units that may be used in accordance with certain embodiments.

[0006] FIG. 2A is an isometric view of a front body portion of a vehicle defining a front bay in accordance with certain embodiments.

[0007] FIG. 2B is an isometric view of the front body portion of the vehicle having a tub and access panel mounted in a front bay of the vehicle in accordance with certain embodiments.

[0008] FIG. 2C is an isometric view of the front body portion of the vehicle having an apron panel mounted in the front bay of the vehicle in accordance with certain embodiments.

[0009] FIGS. 3A and 3B illustrate pivoting of the access panel relative to the tub in accordance with certain embodiments.

[0010] FIGS. 4A, 4B, and 4C illustrate the operation of a living hinge coupling the access panel to the tub in accordance with certain embodiments.

[0011] FIG. 5 is an isometric view showing latching of the access panel with respect to a leaf screen and a dust seal secured around the tub in accordance with certain embodiments.

[0012] FIG. 6 illustrates a gap formed between the tub and the access panel in accordance with certain embodiments.

[0013] FIG. 7 illustrates interaction between the access panel and the dust seal during pivoting in accordance with certain embodiments.

[0014] FIG. 8 is a side view illustrating a fastener for latching the access panel to the leaf screen in accordance with certain embodiments.

DETAILED DESCRIPTION

[0015] A tub used to store cargo (e.g., a frunk) is positioned in a bay defined by a vehicle. The tub is secured to an access panel by a living hinge. The access panel may provide access to, for example, one or more reservoirs for fluids or other components. The tub and access panel are formed monolithically in a common mold. Mounting of the access panel to the tub does not require additional fasteners. Side portions mounted to the tub may be connected to the access panel by portions of the living hinge with the tub being disconnected from the access panel between the side portions. The access panel may releasably latch to a component in the bay, such as a leaf guard.

[0016] FIG. 1A illustrates an example vehicle **100** in which the approach described herein may be implemented. As seen in FIG. 1A, the vehicle **100** has multiple exterior cameras **102** and one or more front displays **104**. Each of these exterior cameras **102** may capture a particular view or perspective on the outside of the vehicle **100**. The images or videos captured by the exterior cameras **102** may then be presented on one or more displays in the vehicle **100**, such as the one or more front displays **104**, for viewing by a driver. The vehicle **100** further includes a plurality of road wheels **105**, such as four, that are driven to propel the vehicle **100** over a surface.

[0017] Referring to FIG. 1B, the vehicle **100** may include a chassis **106** including a frame **108** providing a primary structural member of the vehicle **100**. The frame **108** may be formed of one or more beams or other structural members or may be integrated with the body of the vehicle (e.g., unibody construction).

[0018] In embodiments where the vehicle **100** is a battery electric vehicle (BEV) or possibly a hybrid vehicle, a large battery **110** is mounted to the chassis **106** and may occupy a substantial portion (e.g., at least 80 percent) of an area within the frame **108**. For example, the battery **110** may store from 100 to 200 kilowatt hours (kWh). The battery **110** may be a lithium-ion battery or other type of rechargeable battery. The battery may be substantially planar in shape.

[0019] Power from the battery **110** may be supplied to one or more drive units **112**. Each drive unit **112** may be formed of an electric motor and possibly a gear train providing a gear reduction. In some embodiments, there is a single drive unit **112** driving either the front wheels or the rear wheels of the vehicle **100**. In another embodiment, there are two drive units **112**, each driving either the front wheels or the rear wheels of the vehicle **100**. In yet another embodiment, there are four drive units **112**, each drive unit **112** driving one of four wheels of the vehicle **100**.

[0020] Power from the battery **110** may be supplied to the drive units **112** by one or more sets of power electronics **114**, such as power electronics for each drive unit **112** or pair of drive units **112**. The power electronics **114** may include inverters configured to convert direct current (DC) from the battery **110** into alternating current (AC) supplied to the motors of the drive units **112**. The power electronics **114** further facilitate operation of the motors of the drive units as generators to provide regenerative braking. The power electronics **114** further facilitate the transfer of regenerative current to the battery **110**.

[0021] The drive units **112** are coupled to two or more hubs **116** to which wheels **105** may mount. Each hub **116** includes a corresponding brake **118**, such as the illustrated disc brakes. Each hub **116** is further coupled to the frame **108** by a suspension **120**. The suspension **120** may include metal or pneumatic springs for absorbing impacts. The suspension **120** may be implemented as a pneumatic or hydraulic suspension capable of adjusting a ride height of the chassis **106** relative to a support surface. The suspension **120** may include a damper with the properties of the damper being either fixed or adjustable electronically.

[0022] In the embodiment of FIG. 1B, the vehicle **100** is a battery electric vehicle. However, a hybrid-electric vehicle may also benefit from the approach described herein. In particular, the vehicle **100** may be any vehicle having a front trunk or a rear trunk implemented according to the approach described herein.

[0023] Referring to FIG. 2, the vehicle **100** may include a front body portion **200** mounted to the frame **108** or forming part of the frame **108** of the vehicle **100** (e.g., unibody construction). The front body portion **200** defines the exterior of the vehicle forward of the driver. The front body portion **200** therefore may include one or more pieces of sheet metal, plastic, or other materials forming the exterior of the vehicle **100**. The examples disclosed herein are with respect to a front body portion **200** extending over and around the front wheels **105** of the vehicle **100** with the understanding that a trunk or other accessible cavity in a rearward body portion of the vehicle **100** may be implemented in a like manner.

[0024] The vehicle **100** may be understood with respect to X, Y, and Z directions. The Z direction may correspond to the direction of gravity when the vehicle **100** is on a flat, level surface

perpendicular to the direction of gravity. The Y direction may correspond to the direction of the vehicle **100** when traveling in a straight line. The X direction may be defined as perpendicular to the Y and Z directions.

[0025] The front body portion **200** may include wheel wells **202** for containing the road wheels **105** mounted to the hubs **116**, front quarter panels **204** extending over the wheel wells **202**, fenders **206**, or other components extending around the front wheels of the vehicle **100**. The front body portion may likewise include a bumper **208** and grill **210** extending between the front quarter panels **204**. In an electric vehicle, the grill **210** may simply be a front facing panel or may provide an inlet for cooling air flow. The bumper **208** and/or grill **210** may have components mounted thereon, such as headlights, turn signal lights, a light bar, sensors (ultrasonic, radar, etc.), or other components. The front body portion **200** may extend over a portion of the frame **108** defining a front crumple zone for mitigating harm to passengers during collisions.

[0026] The front body portion **200** defines a bay **212**. The bay **212** may define a volume used for storage, e.g., a front trunk (“frunk”) for storing cargo. The bay **212** may also define a volume for functional components of the vehicle such as reservoirs for fluids (brake fluid, coolant, windshield wiper fluid, etc.), degas container, electronic components (e.g., one or more electronic control units (ECU), or other components of the vehicle **100**. Portions of the suspension **120** may also extend into the bay **212**.

[0027] The bay **212** may be covered by a hood **214**. The hood **214** may be secured by hinges **216** to a structural member mounted to the frame **108**. The hinges **216** may enable pivoting of the hood **214** about an axis substantially (e.g., within 5 degrees of) parallel to the X direction. The hood **214** may be supported by gas springs **218** or other supports to maintain the hood **214** in an open position once moved to the open position (e.g., by a user). When in a closed position, the hood **214** may cooperate with the front quarter panels **204** and grill **210** to cover the bay **212**.

[0028] The bay **212** may include various cosmetic and functional panels that cover functional components mounted within the bay **212**. For example, a leaf guard **220** may extend from a windshield **222** to the bay **212** and possibly into the bay **212**. The leaf guard **220** may conduct water and debris descending from the windshield **222** away from the bay **212** and otherwise block debris from entering the bay **212** through a gap between the windshield **222** and the hood **214** when closed. The leaf guard **220** may define slots **224** for receiving the gas springs **218** when the hood **214** is closed and possibly providing clearance for struts coupling the hood **214** to the hinges **216**.

[0029] Referring specifically to FIG. 2B, the bay **212** may contain a tub **226**. The tub **226** defines a volume that is available to store cargo and is relatively isolated from the remainder of the bay **212** and the environment of the vehicle **100** when the hood **214** is closed. For example, the tub **226** may define an available volume (e.g., available for storing cargo) when the hood **214** closed that is at least 1, 2, or more times the size of carry-on luggage having a size approved by the Transportation Security Administration (TSA), e.g. 9 inches by 14 inches by 22 inches. Stated differently, the tub **226** may define an available volume of at least 1.6 cubic feet or more, such as at least 3.2 cubic feet. The upper edge of the tub **226** may be oriented parallel to or at a slight incline (e.g., less than 10, less than 5, or less than 2 degrees) relative to the plane defined by the X and Y directions. The upper edge may be planar or contoured to conform to an underside of the hood **214**. The upper edge may lie within a 15, 10, or 5 centimeters of a plane that is oriented at a slight incline (e.g., less than 10, less than 5, or less than 2 degrees) relative to the plane defined by the X and Y directions.

[0030] The tub **226** is monolithically formed with an access panel **228** using a common mold, such as by gate sequenced injection molding. Gate sequenced injection molding may include activating a first set of gates to inject a first polymer portion into a mold followed by opening a second set of gates to inject a second polymer portion into the same mold. The access panel **228** substantially covers an area over the bay **212** between the tub **226** and the leaf guard **220** and between the tub **226** and the perimeter of the bay **212**, such as spanning at least 90 percent of a distance from the tub **226** and the perimeter of the bay **212**. For example, the access panel **228** may further extend

outwardly from the tub **226** (e.g., in the X direction) substantially to the front quarter panels **204**. The access panel **228** may extend rearwardly (e.g., in the Z direction) from the tub **226** to the leaf guard **220** and may overlap the leaf guard **220** in the Z direction or a flange or other component secured to the leaf guard **220**.

[0031] Access panel **228** may be substantially planar, such as within 5, 3, or 1 centimeter from planar across at least 60, 80, or 90 percent of the extent thereof in the X and Y directions. The access panel **228** may be oriented parallel to or at a slight incline (e.g., less than 10, less than 5, or less than 2 degrees) relative to the plane defined by the X and Y directions. The access panel may define slots **230** that provide clearance for the gas springs **218** and possibly structural components connecting the hood **214** to the hinges **216**. The slots **230** may be aligned with the slots **224** in the leaf guard **220**.

[0032] The access panel **228** may be joined to the tub **226** by side portions **228a**. The upper edge of the tub **226** may be otherwise detached from the access panel **228**. In particular, the portion of the upper edge of the tub **226** extending between the side portions **228a** is detached from the access panel **228**. The side portions **228a** may secure to a structural member within the bay **212** using fasteners **228b**. The tub **226** and any cargo placed therein may be supported by structural members other than the side portions **228a** such that the side portions **228a** function primarily as a covering to improve aesthetics of the bay **212** and hinder contamination of functional components covered by the side portions **228a**.

[0033] The tub **226** may further be secured to structures within the bay **212**. For example, the tub **226** may include a flange **226a**, such as on an opposite side of the tub **226** from the access panel **228** along the Y direction. Openings **226b** formed in the flange **226a** may receive fasteners securing the flange **226a** to a structure within the bay **212**.

[0034] The leaf guard **220**, apron panel **232**, and the combined tub **226**, access panel **228**, and side portions **228a** may be formed of plastic, metal, or composite material. For example, these components may be formed from any plastic used for forming interior panels, engine covers, or other trim as known in the automotive industry (e.g., nylon, acrylonitrile butadiene styrene (ABS), polypropylene, polyvinyl chloride (PVC), or the like).

[0035] Referring to FIG. 2C, a remaining portion of the bay **212** that is not covered by the tub **226**, access panel **228**, side portions **228a**, and the leaf guard **220** may be covered by an apron panel **232**. The apron panel **232** may be substantially planar, such as within 5, 3, or 1 centimeter from planar across at least 60, 80, or 90 percent of the extent thereof in the X and Y directions. The apron panel **232** may be oriented parallel to or at a slight incline (e.g., less than 10, less than 5, or less than 2 degrees) relative to the plane defined by the X and Y directions.

[0036] The apron panel **232** and access panel **228** may form a contoured surface. For example, a radius of curvature of the apron panel **232** may be substantially (e.g., within 1 percent of) equal to the radius of curvature of the access panel **228**, excluding a region within 1 centimeter of the edges of the apron panel **232** and access panel **228** that are facing one another to account for bevels, chamfers, or other edge treatments of the edges.

[0037] Referring to FIGS. 3A and 3B, material joining the side portions **228a** to the access panel **228** may be flexible, e.g., a living hinge, such that the access panel **228** may pivot relative to the side portions **228a** without failure. For example, the access panel **228** may pivot about an axis of rotation that is substantially (e.g., within 5 degrees of) parallel to the axis of rotation of the hood **214**, such as an axis of rotation substantially parallel to the X direction.

[0038] The access panel **228** therefore has a closed position shown in FIGS. 2B and 2C in which the access panel **228** remains while the hood **214** is closed and in normal operation. The access panel **228** further has an open position shown in FIGS. 3A and 3B that is achievable when the hood **214** is open. The access panel **228** may be pivoted to the open position to fill one or more reservoirs and/or to access components positioned around the tub **226** in the bay **212**.

[0039] As shown in FIGS. 3A and 3B, an access opening **300** may be defined between the tub **226**

and the leaf guard **220**. The access opening **300** may further extend between left and right front quarter panels **204** of the vehicle **100**. The access opening **300** may provide access to some or all of reservoirs for fluids (e.g., brake fluid, coolant, windshield wiper fluid, etc.), degas container, electronic components (e.g., one or more electronic control units (ECU), portions of the suspensions **120**, or other components of the vehicle **100**.

[0040] FIGS. **4A**, **4B**, and **4C** illustrate an example approach for implementing a living hinge attaching the access panel **228** to the side portions **228a**. The approach shown in FIGS. **4A**, **4B**, and **4C** is exemplary only, and other approaches for implementing a living hinge using compliant materials may also be used.

[0041] In some embodiments, a living hinge is implemented by two hinge portions **400**, where each hinge portion **400** joins one of the side portions **228a** to the access panel **228**. For example, the access panel **228**, and possibly each side portion **228a** may have a thickness **T1** whereas the hinge portion **400** has a thickness **T2** that is much less than **T1**, such as less than 0.5, 0.25, or less than 0.15 times the thickness **T1**. For example, **T1** may be between 2 and 4 millimeters, such as between 2.5 and 3.5 mm, such as between 2.9 and 3.1 millimeters. In contrast, **T2** may be between 0.3 and 1 millimeters, such as between 0.35 and 0.75 millimeters, such as between 0.45 and 0.65 millimeters, such as between 0.5 and 0.6 millimeters. For example, **T2** may be about 0.55 millimeters. In some embodiments **T1** and **T2** are measured along the Z direction.

[0042] As is apparent in FIG. **4A**, there may be a sloped or curved transition between the thickness **T1** and the thickness **T2** of the hinge portion **400**. For example, the curved transition may have an extent of between 0.5 and 2 centimeters. An apex of the hinge portion **400** may have a radius of curvature **R** selected to reduce force concentrations and extend the fatigue life of the hinge portion **400**. The apex may be defined as the region having the smallest thickness when the access panel **228** is in the closed position. **R** may be at least 0.1 millimeters, at least 0.2 millimeters, at least 0.4 millimeters, or at least 0.8 millimeters.

[0043] In some embodiments, one or more features may be formed on the side portions **228a** to position the access panel **228** relative to the apron panel **232** in order to present a smooth contour for aesthetic or functional reasons. For example, a flange **402** extending downwardly from the apron panel **232** in the Z direction may rest on the side portions **228a**. The flange **402** may provide a space below the apron panel **232** to provide clearance for the fasteners **228b** covered by the apron panel **232** or for other purposes. A flange **404** may be formed on each side portion **228a**. Each flange **404** extends upwardly from each side portion **228a** in the Z direction and connects to the hinge portion **400** such that the hinge portion **400** joins the flange **404** to the access panel **228**. The flange **404** may maintain the access panel **228** at a height **H** above the side portion **228a**. The flange **402** may be monolithically formed with the apron panel **232**, such as by molding in a common mold, such as using gate sequenced injection molding. The flange **404** may be monolithically formed with the access panel **228** and side portions **228a**, such as by forming in a common mold, such as using gate sequenced injection molding.

[0044] The flange **404** may have a thickness **T3**, such as in the Y direction. The thickness **T3** may be less than the thickness **T1**. For example, **T3** may be between 1 and 3 millimeters, such as between 1.5 and 2.5 millimeters. In some embodiments, the reduced thickness **T3** may facilitate bending of the flange **404** during lifting of the access panel **228** (see FIG. **4B**). For example, at least 5, 10, 20, or 30 percent of an angle traversed by the access panel **228** (e.g., 35 degrees) from the closed position of FIG. **4A** may be a result of bending of the flange **404** with a remainder being a result of deformation of the hinge portion **400**, e.g., a region within 3 millimeters of the apex of the hinge portion **400**.

[0045] However, in other embodiments, the thickness **T3** is selected such that no substantial bending of the flange **404** occurs during rotation of the access panel, e.g., less than 5 percent or less than 2 percent of the angle traversed by the access panel **228** (e.g., 35 degrees) from the closed position of FIG. **4A** may be a result of bending of the flange **404** with a remainder being a result of

deformation of the hinge portion **400**.

[0046] In either case, the access panel **228** has an effective axis of rotation defined by the hinge portions **400** that is substantially (e.g., within 5 degrees of) parallel to the X direction. The location of the axis of rotation in the Y-Z plane may be defined by a point in the hinge portions **400** or offset therefrom where substantial bending of the flange **404** occurs.

[0047] In various embodiments, the access panel **228** is not regularly pivoted during use.

Accordingly, the hinge portion **400** (and the flange **404** if substantial bending occurs) may have a fatigue life selected to avoid failure during the expected service life of the vehicle **100**. For example, the fatigue life may be between 200 and 300 cycles (opening and closing) assuming about 10 uses per year and a service life of at least 20 years. To reduce likelihood of premature failure, the fatigue life may be greater, such as between 300 and 1000 cycles.

[0048] As is apparent in FIGS. **4A** and **4B**, the flanges **402**, **404** may be offset from one another in the Y direction, such as by a distance **T4**. The offset may be selected for aesthetic reasons (perception of quality due to panel gaps) or functional reasons. For example, **T4** may be selected such that, for an expected range of movement of the access panel **228** in the Y direction when moving from the closed position to the open position (e.g., an angle between 30 and 60 degrees), deflection of the flange **404** in the Y direction will not result in contact with the apron panel **232** by the flange **404**, access panel **228**, hinge portion **400**, or any other portion molded with the access panel **228**, the hinge portion **400**, and the flange **404**.

[0049] Referring to FIGS. **5**, **6**, and **7**, a dust seal **500** may extend around an upper perimeter of the tub **226**. When the hood **214** is closed, the dust seal **500** may engage an inner surface of the hood **214**, a corresponding seal secured to the inner surface of the hood **214**, or other structure secured to the hood **214**. The engagement of the dust seal **500** with the hood **214** (directly or by way of another structure secured thereto) resists entry of contaminants into the tub **226**. The dust seal **500** may be made of a natural or synthetic rubber, silicone, or another compliant polymer. The dust seal **500** may be made of a compliant material, such as with a hardness of between 20 and 70 Shore A.

[0050] As seen in FIG. **6**, a gap **600** may be present at an interface between the tub **226** and the access panel **228** to facilitate movement of the access panel **228** relative to the tub **226**. The gap **600** may extend between the side portions **228a** and may extend between the tub **226** and the hinge portion **400** and the flange **404**. The gap **600** therefore extends along the extent of the tub **226** in the X direction and partially along the extent of the tub **226** in the Y direction. Accordingly, the hinge portion **400** and flange **404** may be secured to the tub **226** only by way of the side portions **228a**, thereby facilitating movement of the hinge portion **400** and possibly bending of the flange **404**.

[0051] The gap **600** may be formed after molding of the tub **226**, access panel **228**, and side portions **228a** or may be formed in a subsequent manufacturing step. In some embodiments, the gap **600** is formed by a core out implemented by dies used to mold the tub **226**, access panel **228**, and side portions **228a**.

[0052] Referring specifically to FIG. **7**, the gap **600** may be selected to facilitate free movement of the access panel **228** relative to the tub **226**. For example, the dust seal **500** may be secured to a rib **700** extending around the tub **226**, such as a rib **700** completely around the tub **226**. In the illustrated embodiment, the rib **700** is formed on an outwardly extending flange **702** defining a perimeter of the tub **226**. The access panel **228** may have a corresponding flange **704** extending along the gap **600**.

[0053] When pivoting from the closed position to the open position, the flange **704**, or other portion of the access panel **228**, may be moved closer to the tub **226** in the Y direction. In some implementations, contact between the flange **704**, or other portion of the access panel **228**, and the dust seal **500** may occur. Compliance of the dust seal **500** may limit or prevent damage from such contact. In other embodiments, the gap **600** is sized such that such contact does not occur.

[0054] Referring again to FIG. **5**, when in the closed position, the access panel **228** may latch relative to the remainder of the vehicle **100**, such as by latching to the leaf guard **220** or other

structure within the bay **212**. In the illustrated embodiments, releasable fasteners **502** are non-releasably secured to the access panel **228** and releasably secured to the leaf guard **220**. Alternatively, the fasteners **502** may releasably secured to the access panel and be non-releasably secured to the leaf guard **220**. As used herein, “releasable” may be understood as requiring less than 15, 10, or 5 pounds of force to release and “non-releasable” may be understood as requiring more than 15, 20, or 25 pounds of force to release. The fasteners **502** may releasably latch without the use of tools. In the illustrated embodiment, the fasteners releasably engage extensions **504** formed on the leaf guard **220** that extend under the access panel **228**. Alternatively, the fasteners **502** may be non-releasably secured to the extensions **504** and releasably engage openings defined by the access panel **228**.

[0055] Referring to FIG. **8**, the fasteners **502** may be implemented as any manually releasable fastener known in the art, particularly those used for securing decorative trim or internal panels to a vehicle as known in the automotive industry. For example, the fasteners **502** may be implemented as the illustrated retainer clip **800**. The retainer clip **800** may include a head **802** that non-releasably secures the retainer clip **800** to the access panel **228**, such as within an opening **804** defined by the access panel **228**. In the illustrated embodiment, the opening **804** is defined in a bracket **806** secured to the access panel **228**, referred to as a “dog house” in the automotive industry. The leaf guard **220**, such as an extension **504** secured to the leaf guard **220**, may receive a clip portion **808** of the retention clip **800**. For example, the extension **504** may define an opening **810** into which the clip portion **808** inserts. The clip portion **808** may deflect upon insertion through the opening **810** with a portion thereof biased outwardly following insertion, thereby resisting removal. The clip portion **808** may then be disengaged by pulling upwardly on the access panel **228** with sufficient force to deflect the clip portion **808** and force the clip portion **808** out of the opening **810**. Pulling upwardly on the access panel **228** may be performed manually or using a trim tool.

[0056] The descriptions of the various embodiments of the present disclosure have been presented for purposes of illustration. Many modifications and variations will be apparent to those of ordinary skill in the art without departing from the scope and spirit of the described embodiments. The terminology used herein was chosen to explain the principles of the embodiments, the practical application or technical improvement over technologies found in the marketplace, or to enable others of ordinary skill in the art to understand the embodiments disclosed herein.

[0057] In the preceding, reference is made to embodiments presented in this disclosure. However, the scope of the present disclosure may exceed the specific described embodiments. Instead, any combination of the features and elements, whether related to different embodiments, is contemplated to implement and practice contemplated embodiments. Furthermore, although embodiments disclosed herein may achieve advantages over other possible solutions or over the prior art, the embodiments may achieve some advantages or no particular advantage. Thus, the aspects, features, embodiments and advantages discussed herein are merely illustrative.

[0058] While the foregoing is directed to embodiments of the present disclosure, other and further embodiments may be devised without departing from the basic scope thereof, and the scope thereof is determined by the claims that follow.

Claims

1. A vehicle component comprising: a tub configured to store cargo; an access panel configured to cover at least a portion of a bay defined by a vehicle; and a living hinge coupling the tub to the access panel, wherein the tub, the access panel, and the living hinge are monolithically formed in a common mold.
2. The vehicle component of claim 1, wherein the tub defines a storage volume of at least 1.6 cubic feet.
3. The vehicle component of claim 1, further comprising a dust seal extending around a perimeter

of the tub.

4. The vehicle component of claim 1, further comprising: a first side portion extending outwardly from the tub; and a second side portion extending outwardly from the tub, the tub being positioned between the first side portion and the second side portion, the first side portion and the second side portion being monolithically formed with the tub in the common mold, the living hinge comprising a first hinge portion connecting the first side portion to the access panel and a second hinge portion connecting the second side portion to the access panel.

5. The vehicle component of claim 4, wherein the tub is detached from the access panel between the first side portion and the second side portion.

6. The vehicle component of claim 4, wherein the access panel has a first thickness, the first hinge portion and the second hinge portion have a second thickness at thinnest points thereof, the second thickness being less than 0.25 times the first thickness.

7. The vehicle component of claim 6, further comprising: a first flange extending between the first hinge portion and the first side portion; and a second flange extending between the second hinge portion and the second side portion; wherein the first flange and the second flange have a third thickness that is less than the first thickness and greater than the second thickness.

8. A vehicle comprising: a chassis having one or more drive units and a plurality of wheels mounted thereto; one or more panels defining an exterior of the vehicle mounted to the chassis, the one or more panels defining a bay; a cover having a closed position covering the bay; one or more functional components mounted within the bay; and a molded component positioned in the bay, the molded component comprising: a tub configured to store cargo; an access panel; and a living hinge coupling the tub to the access panel, the access panel being rotatable from a closed position in which the access panel covers the one or more functional components and an open position in which the one or more functional components are accessible.

9. The vehicle of claim 8, wherein the one or more functional components include one or more fluid reservoirs.

10. The vehicle of claim 8, wherein the vehicle comprises a windshield, the access panel being positioned between the windshield and the tub.

11. The vehicle of claim 10, wherein the vehicle comprises a leaf guard positioned at a lower edge of the windshield, the access panel being removably latched to the leaf guard.

12. The vehicle of claim 8, wherein the one or more panels form a grill and left and right front quarter panels of the vehicle.

13. The vehicle of claim 8, further comprising a dust seal extending around a perimeter of the tub, the dust seal positioned to engage the cover when the cover is in the closed position.

14. The vehicle of claim 8, wherein the tub defines a storage volume of at least 1.6 cubic feet.

15. The vehicle of claim 8, wherein the molded component comprises: a first side portion extending outwardly from the tub to a perimeter of the bay; and a second side portion extending outwardly from the tub to the perimeter of the bay, the tub being positioned between the first side portion and the second side portion, the living hinge comprising a first hinge portion connecting the access panel to the first side portion and a second hinge portion connecting the second side portion to the access panel; wherein the tub is detached from the access panel between the first side portion and the second side portion.

16. The vehicle of claim 15, wherein the access panel has a first thickness, the first hinge portion and the second hinge portion have a second thickness at thinnest points thereof, the second thickness being less than 0.25 times the first thickness.

17. The vehicle of claim 16, further comprising: a first flange extending between the first hinge portion and the first side portion; and a second flange extending between the second hinge portion and the second side portion; wherein the first flange and the second flange have a third thickness that is less than the first thickness and greater than the second thickness.

18. A method for accessing a vehicle comprising: opening a cover from a cover closed position to a

cover open position, the cover covering a bay defined by one or more body panels of a vehicle when in the cover closed position; and opening an access panel from a panel closed position to a panel open position, the access panel being part of a molded component including a tub configured to store cargo, the molded component including a living hinge coupling the tub to the access panel, opening the access panel including bending the living hinge; wherein one or more functional components of the vehicle positioned in the bay are exposed when the access panel is open.

19. The method of claim 18, wherein opening the access panel comprises unlatching the access panel from a structure within the bay.

20. The method of claim 18, wherein the molded component comprises: a first side portion extending outwardly from the tub to a perimeter of the bay; and a second side portion extending outwardly from the tub to the perimeter of the bay, the tub being positioned between the first side portion and the second side portion, the living hinge comprising a first hinge portion connecting the access panel to the first side portion and a second hinge portion connecting the second side portion to the access panel; wherein the tub is detached from the access panel between the first side portion and the second side portion.
